

Simulation of Ignition Delay Time of a Syngas Mixture Using Cantera

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Title of Reference Used:

An optimized kinetic model for H₂/CO combustion in CO₂ diluent at elevated pressures

Abstract

The ignition delay parameter is very necessary in understanding the combustion behaviour of alternative fuels such as syngas. The ignition delay time of a syngas mixture was examined in this study using Cantera with an optimised chemical kinetic mechanism. The effect of temperature and pressure (2 atm, 10 atm and 20 atm) under the following syngas mixture: H₂/CO/O₂/CO₂/Ar = 3.5/3.5/3.5/3/211.5 was studied using a zero-dimensional constant-volume reactor. Simulations were performed over temperature ranges from 890 to 1250 K, 960 to 1280 K, and 1000 to 1280 K at the following pressures: 2 atm, 10 atm, and 20 atm, respectively. The results of the simulation show that IDT decreases with increasing temperature and is strongly dependent on pressure, with higher pressures having rapid ignition. The reproduced curves closely aligned with the trends in Figure 11 in the literature and correspond to the expected Arrhenius-type behaviour. There was a successful capture of the overall ignition behaviour of the system by the model with minor deviations, which shows the effectiveness of Cantera in the study of combustion processes and in giving a great insight into syngas behaviour under different conditions.

1 Introduction

The ignition delay time (IDT, τ) can be defined as that period between the creation of a combustible mixture, as by the injection of fuel into an oxidizing environment, and the onset of the rapid reaction phase leading to the rise in temperature and pressure [1]. Reaching a specific temperature, pressure, or species concentration is known as the time point of ignition [1]. Ignition time delay is essential in designing and optimising energy conversion systems such as combustion engines and gas turbines [2].

Synthesis gas (syngas) is a composition of hydrogen (H₂), carbon monoxide (CO), and carbon dioxide (CO₂). It may also include nitrogen utilized for ammonia production [3]. Syngas serves as the principal chemical intermediary within the chemical industry. It is utilized in numerous highly selective syntheses of various compounds and fuels, as well as a source of pure hydrogen and carbon monoxide [4]. Syngas is becoming significantly important in energy conversion.

This work describes the simulation of the ignition delay times of a syngas mixture utilising a Cantera-implemented chemical kinetic model. The main objective is to reproduce the optimised mechanism of Figure 11 in the literature [5].

2 Computing Methods

The mechanism file was downloaded from the supplementary files of the reference paper in Chemkin-format mechanisms. This is converted to a Cantera YAML file using Cantera (version 3.0.1). Cantera is a software toolkit used for chemical kinetics and thermodynamics simulations [6]. Upon conversion, it was displayed that

the mechanism contains 15 species and 47 reactions. In this study, the optimised syngas mechanism with the file name (`optimized_syngas.yaml`) was used as the chemical kinetic mechanism. The simulations were performed using a zero-dimensional constant-volume reactor model [5]. Real gas effect was not considered in the simulation as noted in the reference paper because the effect is minimal compared to measurement uncertainty.

The following were defined for the gas mixture composition:

$$\begin{aligned} \text{H}_2 &= 3.5 \\ \text{CO} &= 3.5 \\ \text{O}_2 &= 3.5 \\ \text{CO}_2 &= 3.0 \\ \text{Ar} &= 211.5 \end{aligned}$$

These are consistent with the conditions set, as seen in the research paper [5].

Over temperature ranges of 890–1250 K, 960–1280 K, and 1000–1280 K at the following pressures: 2 atm, 10 atm, and 20 atm, respectively, ignition delay times were computed. By the application of a reactor network solver, the reactor equations were integrated over time for each condition.

The definition of the ignition delay time was given as the time corresponding to the maximum mass fraction of the OH radical, a commonly used indicator of ignition in combustion studies [7].

3 Results

For the reproduction of Figure 11 of the reference paper, the optimised kinetic mechanism for the three separate

pressures (2, 10, and 20 atm) was used to compute the ignition delay times for the syngas composition. As shown in Figure 1, this result demonstrates a function of inverse temperature on a semilogarithmic scale. Three different curves were obtained for the three pressure conditions. In the cases of the three generated curves, there is a decrease in IDT with temperature increase (when $1000/T$ decreases).

IDT is significantly longer at lower temperatures (that is, when $1000/T$ is at higher values), while there is a sharp decrease in IDT to the values below 10^{-4} at a temperature rise.

A clear pressure dependence is also observed, which is obvious in the obtained three different curves. The IDT is shorter when the pressure is higher at a specified temperature. From the graph, it could be seen that the 20 atm condition has the shortest IDT, 10 atm is intermediate, and then 2 atm has the longest IDT.

The curves show a smooth transition from long ignition delays at low temperatures to shorter ignition delays at higher temperatures. This is in correspondence to the shape and pattern we have in Figure 11 of our reference paper, which shows that our optimized mechanism is in line with the characteristics of the ignition of the syngas system.

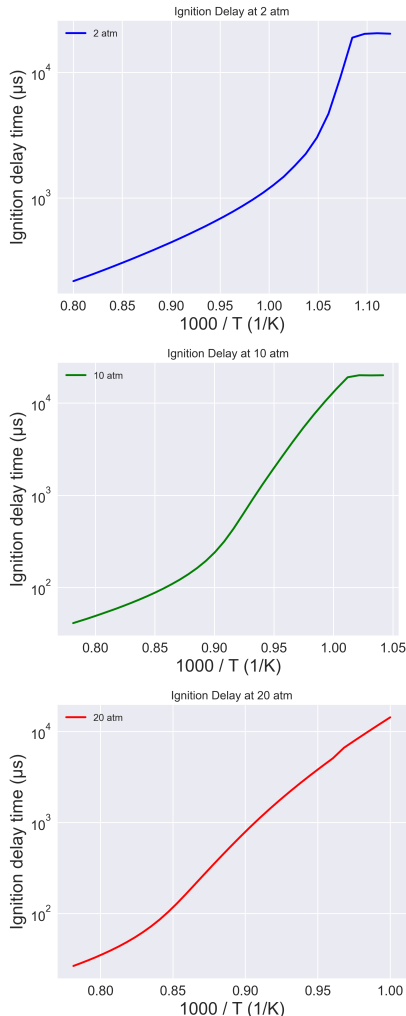


Figure 1: Reproduced figures of the optimised model in Figure 11 of the literature.

4 Analysis

From the obtained results of the simulation, the collective impact of pressure, temperature and chemical kinetics regulating the oxidation of syngas mixtures is reflected by the observed ignition delay behaviour. There is consistency between the trends obtained from the simulation and the fundamental theory of combustion, and this gives an accurate and deep understanding of the involved reaction mechanisms.

4.1 Effect of Temperature

Over all the conditions of pressure, it is clearly seen that IDT depends strongly on temperature. IDT rapidly decreases with an increase in temperature (that is, the decrease in values of $1000/T$), which is a typical pattern of thermally activated reactions, and this goes along with Arrhenius-type kinetics [1].

There is a rapid increase in the rate constants of elementary reactions when the temperature is higher; hence, reactive species like O, H, and OH are being generated faster from this consequence. The produced radicals are very significant in the acceleration of the combustion process due to chain-branching reactions which result in shorter IDT as the system reaches ignition rapidly [2].

The reverse is the case when the temperature is lower because there is a notable reduction of reaction rates as a result of deficient thermal energy to overcome activation barriers; hence, the production of radicals here is slower, leading to prolonged ignition delays [2]. The sensitivity of the ignition delay to temperature changes is stressed by the steep slope observed in the semi-logarithmic plot.

4.2 Effect of Pressure

IDT is also remarkably influenced by pressure. A rapid ignition delay is observed at higher pressures like 20 atm for any given temperature in contrast to lower pressures like 10 atm and 2 atm, which produce longer ignition delays. This pattern is due to the rise in concentration of reactant molecules at higher pressures [8].

Kinetically, the molecular collision frequency increases by the rise in pressure; hence, the chances of successful reactive encounters are raised, and both the chain propagation and chain initiation are accelerated in the reaction mechanism. Furthermore, higher pressures favour pressure-dependent reactions, leading to stabilisation and radical formation [7].

Consequently, a rapid occurrence of ignition at higher pressures is observed. The intense pressure reliance of the ignition process is demonstrated in the plot for different pressures by the separation between the curves, as seen in Figure 2.

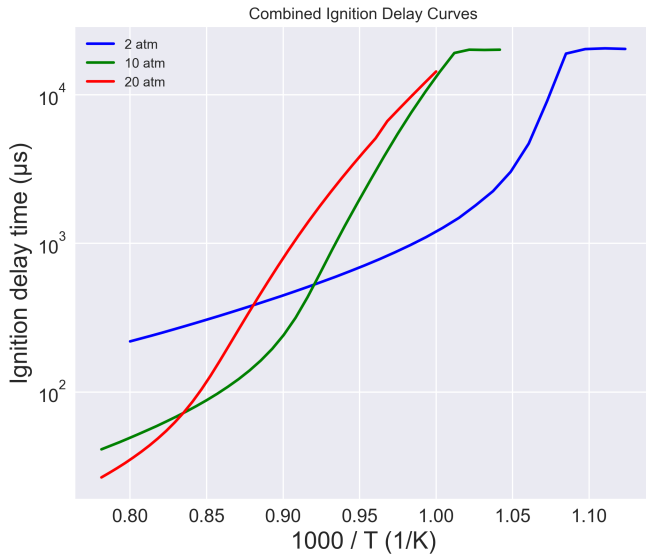


Figure 2: Combined Ignition Delay Curves

4.3 Arrhenius Behavior and Plot Interpretation

The visualisation of the Arrhenius-type behaviour is enabled by plotting the inverse temperature ($1000/T$) on the x-axis. An approximately exponential temperature reliance is shown by IDT on a semi-logarithmic scale. In certain regions of the plot, this appears as a near-linear trend.

Nevertheless, deviation from perfect linearity is observed, especially at the intermediate range of temperature, an absolute suggestion that the ignition process is governed by multiple competing reaction pathways rather than a single dominant reaction [1]. Complex temperature-dependent behaviour in syngas combustion is attributed to the overall kinetics being contributed by both hydrogen and carbon monoxide oxidation pathways [5].

4.4 Chemical Kinetic Considerations

Syngas ignition involves hydrogen oxidation reactions, CO oxidation reactions, and chain-branching reactions involving radicals such as H, O, and OH [5].

OH radical is the key intermediate species, which is utilised as the ignition marker, and the start of ignition is indicated by its speedy increase. The peak in the concentration of OH is in correspondence to the transition to rapid energy release from low-speed pre-ignition chemistry.

The behaviour of the ignition is also influenced by diluents like Ar and CO_2 in the mixture, acting as thermal buffers, and energy is absorbed, and the reactivity of the mixture as a whole is reduced, generating prolonged ignition delays in comparison with undiluted systems [7].

4.5 Comparison with Expected Trends in the Literature (Figure 11)

The overall trends in Figure 11 reproduced by the simulated curves are as follows:

- Reduction in ignition delay with rising temperature
- Reduction in ignition delay with elevated pressure
- Gradual transition between high- and low-temperature regimes

Nevertheless, slight deviation from the reference trends is observed, especially in the region of low temperature where the curves tend to converge. Various factors can cause these deviations, which include the following:

- Dissimilarity in the implemented mechanism and the exact optimized mechanism used in the reference study [5]
- Ignition delay definition based on OH peak, which may differ slightly from other criteria such as maximum temperature rise [9]
- Numerical resolution such as the number of temperature points or time-step size

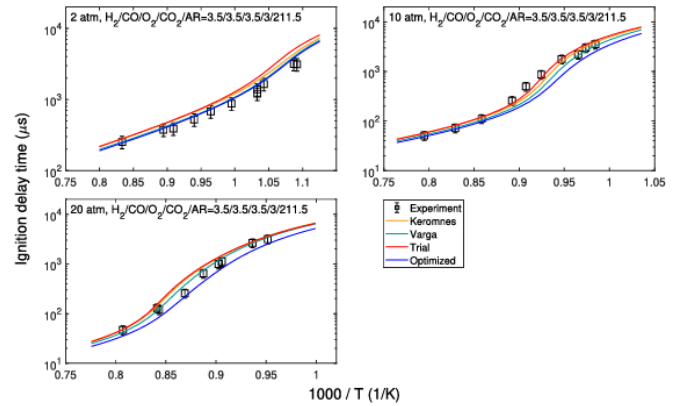


Figure 3: Optimised curves from the reference literature

4.6 Limitations of the Simulation

Here are the recognised limitations, even as the essential trends were illustrated by the model:

- Negligence of spatial effects like convection and diffusion by a zero-dimensional reactor
- The techniques of experimental measurement may not completely or perfectly be represented by defining ignition delay on the basis of only OH concentration [9]
- The consistency of the results depends on the completeness and reliability of the kinetic mechanism [5]

5 Conclusions

In this analysis, optimised chemical kinetic mechanisms implemented in Cantera were effectively utilised to simulate IDT for a syngas mixture. Examination of the pressure and temperature influence on the behaviour of ignition was carried out using a zero-dimensional constant-volume reactor model.

Consistent with Arrhenius-type kinetics, the outcome of the analysis shows that temperature rise decreases the ignition delay while the decline in temperature increases the ignition delay. Furthermore, ignition is accelerated at higher pressures, causing shorter IDT on account of molecular collision frequency increment and improved rate of reaction.

The simulated ignition delay curves reproduce the key qualitative trends observed in Figure 11, confirming the ability of the optimized mechanism to capture the fundamental combustion characteristics of the syngas system. Although minor deviations may exist, the overall agreement indicates that the model provides a reliable representation of ignition phenomena.

The efficacy of computational tools like Cantera in the study of combustion kinetics is illustrated in this work. This work also demonstrates the importance of accuracy in the reaction mechanisms when predicting the behaviour of the ignition. For future investigation, the method used in this work can be applied to other operating conditions and fuel systems.

References

- [1] Su, B., Papp, M., Zhang, P., & Turányi, T. (2024). Dependence of ignition delay time on its definition—A case study on methane ignition. *Combustion and Flame*, 262, 113364.
- [2] Farouk, T. I., Won, S. H., & Dryer, F. L. (2021). Sub-millimeter sized multi-component jet fuel surrogate droplet combustion: Physicochemical preferential vaporization effects. *Proceedings of the Combustion Institute*, 38(2), 3313–3323.
- [3] Rostrup-Nielsen, J., & Christiansen, L. J. (2011). *Concepts in syngas manufacture* (Vol. 10). World Scientific.
- [4] Fiore, M., Magi, V., & Viggiano, A. (2020). Internal combustion engines powered by syngas: A review. *Applied Energy*, 276, 115415.
- [5] Li, W., Zou, C., Yao, H., Lin, Q., Fu, R., & Luo, J. (2022). An optimized kinetic model for H₂/CO combustion in CO₂ diluent at elevated pressures. *Combustion and Flame*, 241, 112093.
- [6] Cantera. Available at: <https://www.cantera.org/>
- [7] Medina, O. E., Gallego, J., Nassar, N. N., Acevedo, S. A., Cortés, F. B., & Franco, C. A. (2020). Thermo-oxidative decomposition behaviors of different sources of n-C7 Asphaltenes under high-pressure conditions. *Energy & Fuels*, 34(7), 8740–8758.
- [8] Shadram, Z., Nguyen, T. M., Sideris, A., & Sirignano, W. A. (2022). Physics-aware neural network flame closure for combustion instability modeling in a single-injector engine. *Combustion and Flame*, 240, 111973.
- [9] Fredrich, D., Jones, W. P., & Marquis, A. J. (2019). The stochastic fields method applied to a partially premixed swirl flame with wall heat transfer. *Combustion and Flame*, 205, 446–456.